UNIT TEST 4 CN

(PHYSICAL, DATA LINK LAYER) , JAVA (CLASS AND OBJECT , INTERFACE)

1. **The portion of physical layer that interfaces with the media access control sublayer is called ____**

(a) Physical signaling sublayer

(b) Physical data sublayer

(c) Physical address sublayer

(d) Physical transport sublayer

2. On what type of switching does internet work?

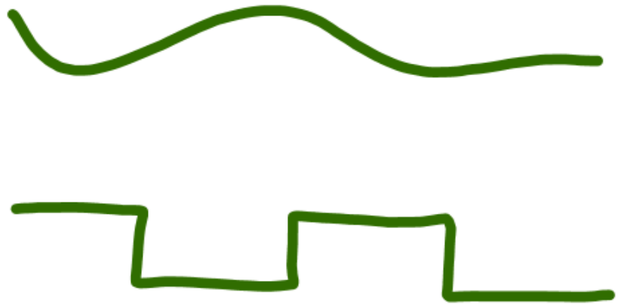
(a) Circuit (b) Bit

(c) Packet (d) Line

3. **Which of the following techniques is used for transmitting digital signal?**

(a) Amplitude modulation

(b) Frequency modulation



(c) Pulse code modulation

(d) Amplitude shift keying

4. **Which of the following is an advantage of computer networks?**

(a) Resource sharing

(b) Improving reliability and availability

(c) Enhancing performance

(d) All of the above

5. In which layer of network architecture, the secured socket layer (SSL) is used?

(a) Physical layer

(b) Session layer

(c) Application layer

(d) Presentation layer

Secure Socket layer → TLS

Application layer

|

Transport layer

6. **X. 21 is**

(a) A method of determining which device has access to the transmission medium at any time

(b) A method access control technique or multiple access transmission media

(c) Very common but oriented at linked protocol issued to ISO

(d) A network access standard for connecting stations to a circuit switch network

7. In early ARPANET, which routing method was used:

(a) Distance vector routing - Local Database

(b) Link state routing → Global Database

(c) Least cost routing

(d) Flooding

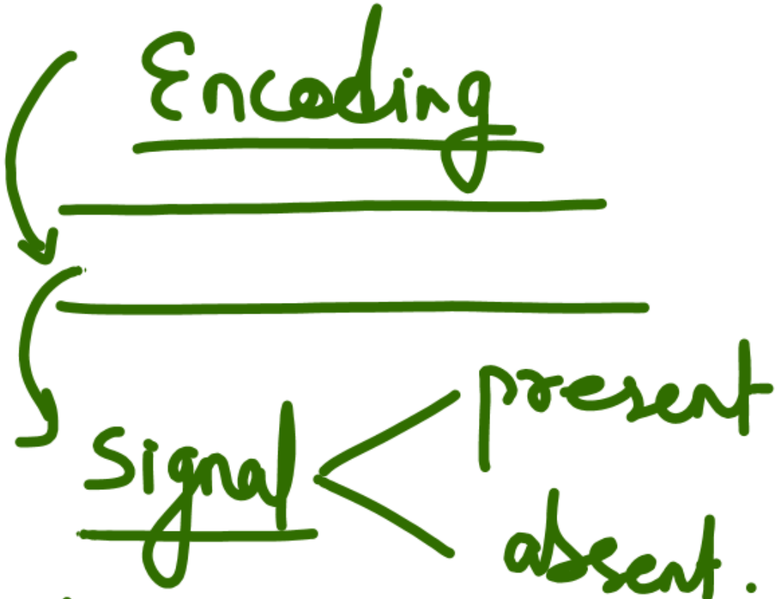
8. Which of the 'following Ethernet encoding has the highest redundancy'

(a) 1B2B

(b) 4B5B

(c) 5B6B

(d) 8B10B



→ optical fiber N/w

→ 8 bit Data → 10 bit Data के रूप में Convert

9. **Which of the following error detection techniques is based on binary division?**

(a) Checksum

(b) Cyclic Redundancy Check (CRC)

(c) Simple parity check

(d) Two-dimensional parity check

10. The term UPnP protocol is a short form for:

(a) Uniform Plug and Play

(b) Unit Plug and Play

(c) Useful Plug and Play

(d) Universal Plug and Play

11. **Which of this is not a network edge device?**

(a) PC

(b) Server

(c) Smartphone

(d) Switch

12. The ____ is the physical path over which a message travels.

(a) Path

(b) Protocol

(c) Route

(d) Medium

13. The structure of format of data is called:

(a) Syntax

(b) Struct

(c) Semantic

(d) None of the above

14. Consider the following statements about the functionality of an IP based router.

I. A router does not modify the IP packets during forwarding.

II. It is not necessary for a router to implement any routing protocol.

III. A router should reassemble IP fragments if the MTU of the outgoing link is larger than the size of the incoming IP packet.

Which of the above statements is/are TRUE?

(a) I and II only (b) I only

(c) II and III only (d) II only

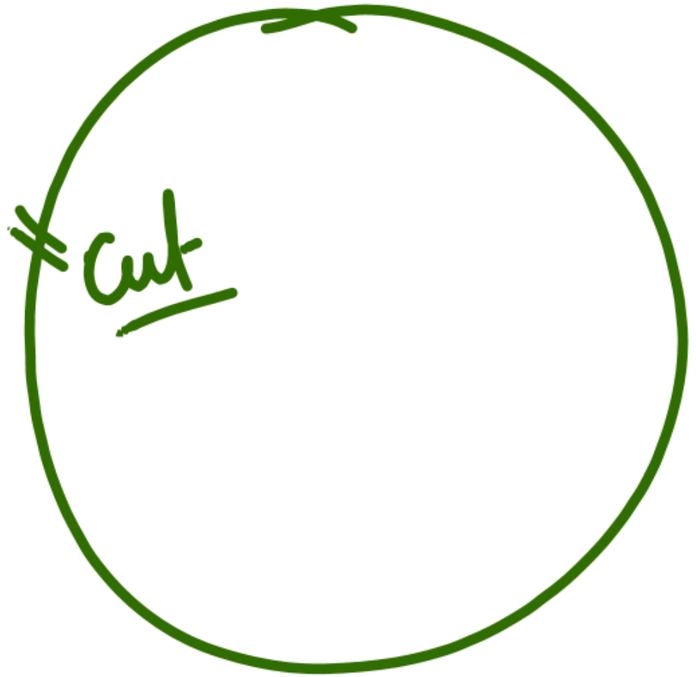
15. If a computer network cable is broken whole network goes down in _____ topology

(a) Star

(b) Ring

(c) Mesh

(d) All options are correct



16. **Hierarchical topology is used to establish a network which is also known as**

(a) Bus network (b) Star network

(c) Tree network (d) Mesh network

17. Which of the following component is responsible for network communication in a computer system?

(a) UPS

(b) Ethernet card NIC

(c) SVGA card

(d) SMPS

18. Speed of internet connection is measured in

(a) Kilo bits per second

(b) Kilo Boolean per second

(c) Meter per second

(d) None of these

19. TCP/IP is an example of

(a) Communication protocol

(b) Networking protocol

(c) Data transfer protocol

(d) All options are correct

20. In a computer network, which of the following is a connecting device?

(a) Router

(b) Hub

(c) Switch

(d) All options are correct

21. _____ is a device that combines transmission from various I/O devices into one line

(a) Modifier (b) Multiplexer

(c) Concentrator (d) Gateway

22. The amount of data transmitted in a fixed amount of time is called

(a) Refresh rate (b) Resolution

(c) Bandwidth (d) Frequency

23. File server is a _____ which is used to store files received from various client machines

(a) Specialized computer system

(b) Modem

(c) Router

(d) Transponder

24. Which of the following is a recovery technique?

(a) Public key encryption

(b) Shadow paging

(c) Hash file organization

(d) Indexed file organization

25. Bit stuffing refers to

(a) Inserting a '0' in user stream to differentiate it with a flag

(b) Inserting a '0' in flag stream to avoid ambiguity

(c) Appending a nibble to the flag sequence

(d) Appending a nibble to the user data stream

26. Dynamic routing protocol enable routers to:

(a) Dynamically discover and maintain router

(b) Distribute routing updates to other routers

(c) Reach agreement with other routers about the network topology

(d) All of the above

27. In Ethernet CSMA/CD, the special bit sequence is transmitted by media access management to handle collision is called

(a) Preamble (b) Post amble

(c) Jam (d) None of these

Jamming Signal

28. Lightweight Directory Access Protocol is used for

(a) Routing the packets

(b) Authentication

(c) Obtaining IP address

(d) Domain name resolving

29. The encoding technique used to transmit the signal in giga Ethernet technology over fiber optic medium is

(a) Differential Manchester encoding

(b) Non Return to zero

(c) 4B/5B encoding

(d) 8B/10B encoding

30. What is the primary purpose of a VLAN?

(a) Demonstrating the proper layout for network

(b) Simulating a network

(c) To create a virtual private network

(d) Segmenting a network inside a switch or device

31. What is the result of compiling and executing the following class ?

```
class AAA
{
   final int a=20;
  public static void main(String[] java)
  {
 int b=30;
 System.out.println(a+b);
  }
}
```

(a) It does not compile

(b) It compile but throws an exception at runtime

(c) It compile and output 50

(d) It compile and output 30

32. what will be the output of following Java code ?

```
class AAA {

    {
        System.out.print("1");
    }

    static
    {
        System.out.print("2");
    }

    AAA()
    {
        System.out.print("3");
    }

    public static void main(String[] java)
    {
        new AAA();new AAA();
    }
}
```

(a) 2133 (b) 21313 (c) 2313 (d) 23131

33. A blueprint for a software object is called

(a) Object

(b) Class

(c) Instance

(d) Encapsulation

34. What will be get output of the following Java code ?

```
protected class AAA {
   public static void main(String[] java)
  {
   String Name="sunil";
  Name=Name+Name;
  System.out.println(Name);
  }
}
```

(a) sunilsunil

(b) sunil

(c) run time exception

(d) compile time error

35. Which of the following statement about a default branch in a switch statement is correct ?

(a) all switch statements must include a default statement

(b) the default statement is required to be placed after all case statements

(c) unlike a case statement the default statement does not take a value

(d) a default statements can only be used when at least one case statement is present

36. what will be the output of the following Java code ?

```
class AAA {
 public static void main(String[] java)
        {
         B obj=null;
      System.out.println(obj.a);
        }
}
class B
{
       int a;
}
```

(a) 0
(b) Garbage value
(c) Compile time error
(d) Run time exception

37. What is the use of ‘this’ keyword in Java?

(a) used to hold the reference of the current object.

(b) holds object value

(c) used to create a new instance

(d) all of these

38. what will be the output of the following java code ?

```
class AAA {
 public static void main(String[] java)
{
 new B();
}}
class B
{
 B()
{
 call();
}
void call()
 {
System.out.println("method callled");}}
```

(a) Compile error bcoz we cannot call method within constructor
(b) method called
(c) run time exception
(d) none of these

39. What is the output of the below Java program with constructors?

```
public class Constructor2
{
        int count=10;
        Constructor2(int count)
        {
    System.out.println("Count=" + count);
         }
           public static void main(String[] args)
         {
    Constructor2 con = new Constructor2();
         }
}
```

(a) Count=0

(b) Count=10

(c) Compiler error

(d) None of the above

40. A constructor can call another overloaded constructor using the ___ keyword in Java.

(a) super

(b) local

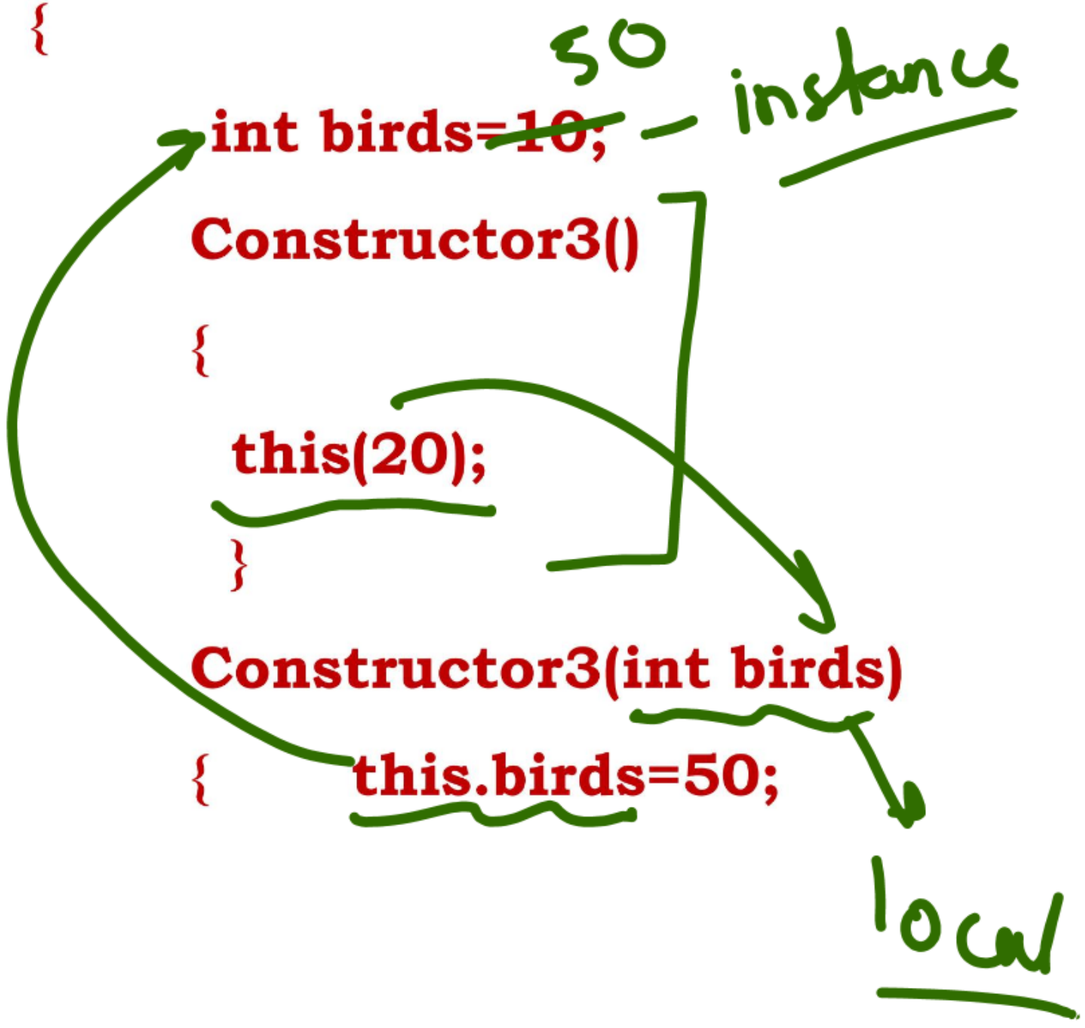
(c) con

(d) this

41. what will me the output of the following java code ?

```
class Constructor3
{
    int birds=10;
    Constructor3()
    {
      this(20);
    }
    Constructor3(int birds)
    {
        this.birds=50;
        System.out.println("Birds=" +birds);
    }

    public static void main(String[] args)
    {
Constructor3 con = new Constructor3();
    }}
```



a) Birds=0
b) Birds=10
c) Birds=20
d) Compile time error

42. In Java, you can pass __ variables from one constructor to another overloaded constructor.

(a) local variables

(b) static variables

(c) non-static variables

(d) local and static variables

static int a = 10

this(a, p)

43. Choosing a suitable overloaded constructor happens at ___ time in Java.

(a) Compile-time

(b) Run time

(c) Depends on Compiler

(d) None of the above

44. A "this" operator used inside a Java method refers to ___ variable.

(a) Global variable

(b) Method local variable

(c) Instance variable

(d) None of the above

45. In Java, local variables are stored in __ memory and instance variables are stored in ___ memory.

(a) Stack, Stack

(b) Heap, Heap

(c) Stack, Heap

(d) Heap, Stack

```
void call ( )
int a = 0;
```

46. What is the output of the below Java program?

```
class Road
{
  static void show()
        {
 System.out.println("Inside static method.");
        }
}

public class TestingMethods10
{
  public static void main(String[] args)
        {
          Road.show();
  } }
```

a) Inside static method.
b) empty message
c) Compiler error
d) Runtime error / exception

47. A Java reference is comparable to ___ in C language.

(a) Enum

(b) Structure

(c) Pointer

(d) None of the above

48. Choose the correct statements about choosing a name for a class in Java.

(a) The class name can start with only a letter or underscore or dollar sign.

(b) The class name can contain numbers

(c) The class name can not start with a number

(d) All of the above

49. In a .java file, how many numbers of public types namely class, interface or abstract can be managed?

(a) 1

(b) 2

(c) 3

(d) Any number

50. Which is true about an anonymous inner class?

(a) It can extend exactly one class and implement exactly one interface.

(b) It can extend exactly one class and can implement multiple interfaces.

(c) It can extend exactly one class or implement exactly one interface.

(d) It can implement multiple interfaces regardless of whether it also extends a class.

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